

W6 Question Bank

Kubernetes Fundamentals

1. What problem does Kubernetes solve? Why do we need container orchestration?
2. Explain the difference between a container and a virtual machine.
3. What are the main differences between Docker Swarm and Kubernetes?
4. In which scenarios would you choose NOT to use Kubernetes?
5. Draw and explain the high-level architecture of a Kubernetes cluster (master/control plane vs worker nodes).
6. What are the key components of the Kubernetes control plane? Explain the role of each.
7. What is etcd? Why is it critical and how is it usually deployed in production?
8. Explain the difference between a Pod, a ReplicaSet, and a Deployment.
9. What is a Pod? Why does Kubernetes use Pods instead of running containers directly?
10. Can a Pod have multiple containers? When would you put multiple containers in one Pod?
11. What happens to a Pod when a node fails?
12. Explain the Pod lifecycle (Pending → Running → Succeeded/Failed).
13. What is kubelet? What is the difference between kubelet and kube-proxy?
14. What is kube-proxy? Explain the three proxy modes (userspace, iptables, IPVS).
15. How does Kubernetes handle networking? What is CNI and name a few popular CNI plugins.
16. What is a Service in Kubernetes? Explain ClusterIP, NodePort, and LoadBalancer types.
17. How does service discovery and DNS work inside a Kubernetes cluster?
18. What is an Ingress? How is it different from a LoadBalancer Service?
19. What is the difference between managed Kubernetes (EKS, GKE, AKS) and self-hosted Kubernetes?
20. What are the benefits of using Amazon EKS over running your own Kubernetes on EC2?
21. How does EKS integrate with other AWS services (IAM, VPC, ALB, CloudWatch, etc.)?

Kubernetes Application Management

22. Explain the difference between Rolling Update and Recreate strategy in Deployments.
23. What is a rolling update's maxSurge and maxUnavailable? How do you choose good values?
24. How do you perform a rollback of a Deployment?
25. What is Horizontal Pod Autoscaler (HPA)? How does it work and what metrics can it scale on?
26. When would Vertical Pod Autoscaler or Cluster Autoscaler be more appropriate than HPA?
27. What is the difference between ConfigMap and Secret?
28. You have a database password — should you use ConfigMap or Secret? Why?
29. How do you update a ConfigMap or Secret without restarting Pods?
30. Explain the different ways to inject environment variables into a Pod.
31. What are PersistentVolume (PV), PersistentVolumeClaim (PVC), and StorageClass?
32. What is dynamic provisioning of storage? How does it work with StorageClasses?
33. Explain the Kubernetes networking model (every Pod gets an IP, Pods can communicate across nodes, etc.). Why is this model important?

34. What is a NetworkPolicy? Give an example of when you would use it.
 35. What does an Ingress Controller do? Name two popular Ingress controllers and explain the difference between them (NGINX Ingress vs ALB Ingress on EKS).
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Monitoring & Observability Stack

1. What is the difference between monitoring and observability? Why has the industry moved toward the term “observability”?
2. Explain the three pillars of observability. In a containerized application, what generates each pillar?
3. Draw and explain the high-level architecture of Prometheus. What are its main components (Prometheus server, Pushgateway, Alertmanager, exporters)?
4. What is the Pull vs Push model in metrics collection? Why did Prometheus choose Pull?
5. What are the four types of Prometheus metrics (Counter, Gauge, Histogram, Summary)? Give a real-world example for each.
6. Write a simple PromQL query to calculate the 95th percentile latency over the last 5 minutes.
7. What is an Exporter? Name at least 4 commonly used exporters in a Kubernetes environment.
8. Explain service discovery in Prometheus. How does it typically work in Kubernetes?
9. What is Alertmanager? How does it handle alert grouping, inhibition, and silencing?
10. Compare Grafana with AWS CloudWatch dashboards — when would you choose one over the other?
11. What is Application Performance Monitoring (APM)? How is it different from traditional infrastructure monitoring?
12. Explain the RED method (Rate, Errors, Duration) and the USE method (Utilization, Saturation, Errors). When do you use each?
13. What is distributed tracing? Explain the concepts of Trace, Span, and Trace Context propagation.
14. Name three popular open-source APM/tracing tools and compare them briefly (Jaeger, Zipkin, Tempo, OpenTelemetry).
15. How do CloudWatch Metrics, CloudWatch Logs, and CloudWatch Alarms differ? Give examples of when you'd use each.

Logging & DevSecOps

16. Why is centralized logging important in microservices and Kubernetes environments?

17. Explain the difference between unstructured and structured logging. Why is structured logging strongly recommended today?
18. Compare Elasticsearch vs OpenSearch — why did AWS fork Elasticsearch into OpenSearch?
19. What are the typical components of the ELK/EFK stack? What is the role of Filebeat/Fluent Bit/Fluentd?
20. How do you manage log retention and control costs in Elasticsearch or CloudWatch Logs in production?
21. What is the difference between SAST, DAST, IAST, and SCA? Give one popular tool for each category.
22. Explain the concept of “Shift-Left Security”. How can you implement it in a CI/CD pipeline?
23. Your application needs to connect to a database. Where and how should you store the database password securely in a modern pipeline? Compare the main options.
24. What are the risks of committing secrets to Git? Name at least three modern tools/strategies to avoid this.
25. What is DevSecOps? How does it extend traditional DevOps practices, and why is it becoming mandatory for most organizations?